

VLSI Internship – Task 1 Report

1. Introduction to VLSI

VLSI (Very Large Scale Integration) is the process of integrating millions of transistors into a single chip. It is used in processors, memory chips, and embedded systems.

2. VLSI Design Flow

- System Specification
- Architecture Design
- RTL Design
- Functional Verification
- Synthesis
- Physical Design
- Fabrication
- Testing

3. Logic Gates Implementation

Basic logic gates were implemented and verified using Logisim.

4. Truth Table - AND Gate

A	B	OUT
0	0	0
0	1	0
1	0	0
1	1	1

5. Truth Table - XOR Gate

A	B	OUT
0	0	0
0	1	1
1	0	1

1	1	0
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6. Half Adder

Sum = A XOR B

Carry = A AND B

A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

7. Testbench (Verification)

All input combinations were applied manually using toggle inputs in Logisim. Outputs were verified against expected truth tables for each logic gate and half adder.

8. Observations

- AND gate outputs 1 only when both inputs are 1.
- XOR gate is used for sum operation.
- Half adder correctly produces sum and carry outputs.